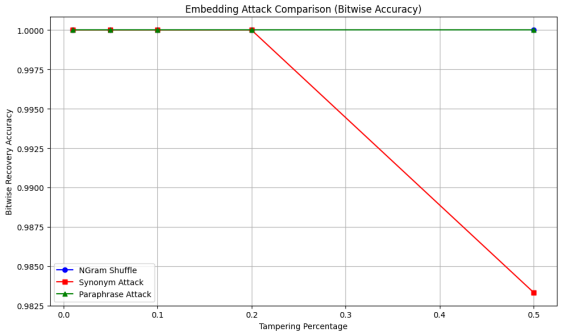
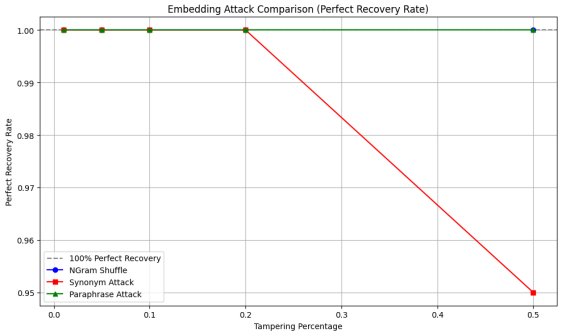


(a) We measure the percentage of bits that are correctly recovered. For example, if [1, 0, 1] was hidden and [0, 0, 1] was recovered, this would contribute a score of 0.67 for that message + attack data point.



(b) We measure whether the bits that are recovered perfectly match the hidden message. For example, if [1, 0, 1] was hidden and [0, 0, 1] was recovered, this would contribute a score of 0 for that message + attack data point.

Fig. 9: We run n-gram shuffle, synonym, and paraphrasing attacks against 20 stegotexts containing 3 bit messages created via the embedding scheme. Attacks are run on each message encoding *hash_output* bits (closer to local mode) at various probabilities. The embedding system resists these attacks at a far stronger rate than the watermarking system, but does not possess the property of "spreading" the hidden bits throughout the entire text.

Overall, our results highlight that embedding-based steganography provides a more practical and resilient method for covert communication, particularly in environments where adversaries employ sophisticated paraphrasing attacks.

8 Applications

In this section, we outline several possible use cases for this type of system.

8.1 Communication Between Citizens Planning Protests

Several works have explored communication in unusual settings, such as environments with no internet access [18,29,7,25,27], environments where people must use government mandated encryption where the government has access to the corresponding secret keys [17], or environments where the adversary can use algorithm-substitution attacks to change users' encryption schemes without users' knowing [5]. This work is suited to the second environment, where the adversary can essentially view all plaintext sent by users. This work further expands Horel et al. allowing the covertext to remain as English text, therefore not raising suspicion as easily and is robust to rerandomization (i.e. paraphrasing) attacks. This defeats the main attack adversaries had against similar systems, allowing for improved security for users.

8.2 Border Crossings & Cloud Storage

Tyagi et al. [36] explores another unusual area with users with sensative needs– dissidents and journalists at border crossings. These users possess highly sensitive data and face dire consequences if apprehended or captured. Tyagi et al. provide a solution through a new file system that provides self-revocable encryption, allowing users to temporarily revoke access to select sensitive files while leaving "normal looking" data intact on their local machine or in the cloud. Our system can be used for a similar purpose by storing sensitive files in collections of covertext. For example, sensitive information can be encoded into a series of short stories or poems. When searched, the government would see the innocent files saved to the device or cloud, along with collections of innocuous writing– which does not seem out of place for someone whose primary job is writing. After the inspection, the hidden data within the poems and short stories can be recovered, restoring the original files. One added benefit is that there is no need to use special storage systems like BurnBox, which may in and of itself raise suspicion. For many realistic adversaries, this may be sufficient to result in punitive action.

9 Conclusion

We introduced formal definitions of weak and strong robust steganography, along with threat models that justify their necessity. To address these challenges, we designed and implemented steganographic schemes that enable users to hide arbitrary secret messages within English covertext generated by large language models. These schemes ensure message recoverability even under adversarial paraphrasing and rewording attacks, significantly advancing the robustness of text-based steganography. Our work provides both theoretical guarantees and practical implementations, demonstrating the feasibility of robust steganographic communication in real-world settings. By open-sourcing our stegosystems and datasets, we aim to facilitate adoption by individuals in need while encouraging further research on key components, such as embedding-based locally sensitive hash functions. We believe this research lays a foundation for more resilient and accessible steganographic methods, contributing to secure and censorship-resistant communication.

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References

1. S. Aaronson. Sign in watermarking of large language models.
2. M. Backes and C. Cachin. Public-key steganography with active attacks. In *Theory of Cryptography Conference*, pages 210–226. Springer, 2005.
3. S. Baluja. Hiding images in plain sight: Deep steganography. *Advances in neural information processing systems*, 30, 2017.
4. M. Bellare, A. Boldyreva, and A. O'Neill. Deterministic and efficiently searchable encryption. In *Advances in Cryptology-CRYPTO 2007: 27th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 19-23, 2007. Proceedings 27*, pages 535–552. Springer, 2007.
5. M. Bellare, K. G. Paterson, and P. Rogaway. Security of symmetric encryption against mass surveillance. In *Advances in Cryptology–CRYPTO 2014: 34th Annual Cryptology Conference, Santa Barbara, CA, USA, August 17-21, 2014, Proceedings, Part I 34*, pages 1–19. Springer, 2014.
6. K. Bennett. Linguistic steganography: Survey, analysis, and robustness concerns for hiding information in text. *CERIAS Tech Report 2004-13*, 2004.
7. A. Bienstock, P. Rösler, and Y. Tang. Asmesh: Anonymous and secure messaging in mesh networks using stronger, anonymous double ratchet. In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*, pages 1–15, 2023.
8. Z. Brakerski and G. Segev. Better security for deterministic public-key encryption: The auxiliary-input setting. In *Advances in Cryptology–CRYPTO 2011: 31st Annual Cryptology Conference, Santa Barbara, CA, USA, August 14-18, 2011. Proceedings 31*, pages 543–560. Springer, 2011.
9. C. Cachin. An information-theoretic model for steganography. In *International Workshop on Information Hiding*, pages 306–318. Springer, 1998.
10. A. Chakraborti, D. Suciu, and R. Sion. Wink: deniable secure messaging. In *32nd USENIX security symposium (USENIX Security 23)*, pages 1271–1288, 2023.
11. M. Chapman and G. Davida. Hiding the hidden: A software system for concealing ciphertext as innocuous text. In *International Conference on Information and Communications Security*, pages 335–345. Springer, 1997.
12. A. Cheddad, J. Condell, K. Curran, and P. Mc Kevitt. Digital image steganography: Survey and analysis of current methods. *Signal processing*, 90(3):727–752, 2010.
13. M. Christ and S. Gunn. Pseudorandom error-correcting codes. In *Annual International Cryptology Conference*, pages 325–347. Springer, 2024.
14. J. Ding, K. Chen, Y. Wang, N. Zhao, W. Zhang, and N. Yu. Discop: Provably secure steganography in practice based on" distribution copies". In *2023 IEEE Symposium on Security and Privacy (SP)*, pages 2238–2255. IEEE, 2023.
15. N. J. Hopper, J. Langford, and L. Von Ahn. Provably secure steganography. In *Annual International Cryptology Conference*, pages 77–92. Springer, 2002.
16. T. Horel, S. Park, S. Richelson, and V. Vaikuntanathan. How to subvert backdoored encryption: Security against adversaries that decrypt all ciphertexts. Cryptology ePrint Archive, Paper 2018/212, 2018. https://eprint.iacr.org/2018/212.
17. T. Horel, S. Park, S. Richelson, and V. Vaikuntanathan. How to subvert backdoored encryption: Security against adversaries that decrypt all ciphertexts. *arXiv preprint arXiv:1802.07381*, 2018.
18. D. Inyangson, S. Radway, T. M. Jois, N. Fazio, and J. Mickens. Amigo: Secure group mesh messaging in realistic protest settings. *Cryptology ePrint Archive*, 2024.
19. N. F. Johnson and S. Jajodia. Exploring steganography: Seeing the unseen. *Computer*, 31(2):26–34, 1998.
20. D. Kahn. The history of steganography. In R. Anderson, editor, *Information Hiding*, pages 1–5, Berlin, Heidelberg, 1996. Springer Berlin Heidelberg.
21. G. Kaptchuk, T. M. Jois, M. Green, and A. Rubin. Meteor: Cryptographically secure steganography for realistic distributions. Cryptology ePrint Archive, Paper 2021/686, 2021. https://eprint.iacr.org/2021/686.
22. A. Karpathy. Karpathy/nanogpt: The simplest, fastest repository for training/finetuning medium-sized gpts.